Idempotent Cayley Graph and its Basic Properties

Dr. Madhavi Levaku[1], Mr. Seetaka Anand[2] and Ms. Ramya Thejeswini[3]

[1] *Professor, Department of Applied Mathematics, Yogi Vemana University, kadapa-516005. Email:* lmadhaviyvu@gmail.com

[2,3] *Research Scholars, Department of Applied Mathematics, Yogi Vemana University, kadapa-516005. Email:* anandkumarcmd2000@gmail.com; kramya8008@gmail.com

Abstract:

The idempotent Cayley graph of the ring $(Z_n,\oplus,\odot)$ is the Cayley graph associated with the symmetric set consisting of idempotent elements and their inverses in the group $(Z_n,\oplus)$. In this paper, we derive some basic properties of the idempotent elements and construct the idempotent Cayley graph. It is shown that this graph is connected and Hamiltonian. Further, if $n = 2^{\alpha}, \alpha > 1$, this graph is bipartite.

Keywords: Idempotent element, idempotent Cayley graph, connected, bipartite, Hamiltonian.